

Satellite radial-profile systematics in DESI LRG HOD inference from public Abacus full-box mocks

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ABSTRACT

Halo-occupation analyses of luminous red galaxy (LRG) clustering usually infer the satellite fraction and satellite velocity bias using high-fidelity mock catalogues. We present a resource-limited but full-volume pilot analysis designed to isolate which part of the modelling drives the inferred satellite fraction. Using the public AbacusSummit base simulation at $z = 0.8$, a compact halo/subhalo-like candidate catalogue, and a DESI DR1 LRG clustering data vector over $0.6 < z < 0.8$, we fit exact-number-density SHAM-like and minimal-HOD samples to $w_p(r_p) + \xi_0 + \xi_2$. The baseline minimal-HOD grid favours a low satellite fraction, $f_{\text{sat}} = 0.04$, with $\alpha_s = 1.1$, while the SHAM-like construction prefers still lower satellite fractions. Observable splits show that this preference is driven primarily by small-scale w_p . A phenomenological radial-dilution test, in which satellite offsets from host centres are scaled by q , improves the heuristic fit from $\chi^2 = 282.6$ to $\chi^2 = 222.3$ and shifts the best point to $f_{\text{sat}} = 0.06$ and $q = 4.0$. However, neither this test nor an NFW concentration-remapping control recovers the literature-scale $f_{\text{sat}} \simeq 0.10$ – 0.15 while simultaneously matching the redshift-space multipoles. This is not a replacement for full DESI AbacusHOD inference, but a diagnostic study of satellite-profile systematics. The result highlights satellite radial profiles, satellite selection, and velocity bias as a coupled systematic in LRG HOD inference.

Key words: large-scale structure of Universe – galaxies: haloes – galaxies: statistics – cosmology: observations – methods: numerical

1 INTRODUCTION

Galaxy redshift surveys use biased tracers to constrain the expansion history and the growth of structure. BAO measurements are mainly protected by the large acoustic scale, whereas RSDs depend directly on peculiar velocities and therefore on how observed galaxies occupy dark-matter haloes (Kaiser 1987; Hamilton 1998). This is especially relevant for DESI LRGs, which provide a high-bias tracer across the transition from low-redshift galaxy surveys to higher-redshift samples (DESI Collaboration et al. 2016; Adame et al. 2025; DESI Collaboration et al. 2024).

The halo occupation distribution (HOD) describes the central and satellite content of haloes, while SHAM assigns galaxies by ranking halo or subhalo properties (Peacock & Smith 2000; Zheng et al. 2005; Conroy et al. 2006; Reddick et al. 2013). For LRGs the satellite fraction is modest, but satellites still control small-scale $w_p(r_p)$ and Fingers-of-God damping, coupling the inferred satellite fraction to velocity bias and the satellite radial profile (Tinker 2007; Guo et al. 2015; Navarro et al. 1996). Thus a simplified model with a mismatched satellite reservoir can be driven toward a biased f_{sat} even if the large-scale multipoles favour a different region of parameter space.

Full DESI-era analyses address this with high-fidelity simulations and extended HOD models. AbacusSummit supplies the large-

volume N -body backbone (Maksimova et al. 2021), and AbacusHOD includes richer occupations, velocity bias and survey-relevant mock construction (Yuan et al. 2022). DESI One-Percent applications find LRG satellite fractions of order ten per cent (Yuan et al. 2024). Here we ask a narrower diagnostic question: using only public full-box Abacus halo and satellite information, which observable drives the low- f_{sat} preference in a simplified HOD/SHAM fit? We fit a DESI DR1 LRG $w_p + \xi_0 + \xi_2$ vector at $0.6 < z < 0.8$, compare SHAM-like and minimal-HOD occupations, and add phenomenological and NFW-based satellite-profile controls. The aim is not a final DESI HOD inference, but to determine whether a low inferred f_{sat} is a physical statement or a symptom of one-halo profile mismatch.

2 DATA AND SIMULATIONS

2.1 Observed clustering vector

We use a DESI DR1 LRG clustering vector over $0.6 < z < 0.8$, combining North and South Galactic caps. The vector consists of projected clustering $w_p(r_p)$ over $0.5 < r_p < 30 h^{-1}\text{Mpc}$ and redshift-space multipoles ξ_0, ξ_2 over $5 < s < 60 h^{-1}\text{Mpc}$. The number density used for exact matching is

$$\bar{n} = 5.31589 \times 10^{-4} h^3 \text{Mpc}^{-3}, \quad (1)$$

corresponding to 4.252712×10^6 galaxies in a $(2000 h^{-1}\text{Mpc})^3$ periodic box. For covariance-weighted observable splits we use a 96-

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region k-means jackknife covariance matched to the same $w_p + \xi_0 + \xi_2$ vector, with the usual Hartlap factor applied to each split.

2.2 Abacus full-box candidate catalogue

The simulation backbone is AbacusSummit_base_c000_ph000 at $z = 0.8$ (Maksimova et al. 2021). We use the full $2000 h^{-1} \text{Mpc}$ periodic box and a compact candidate catalogue built from `halo_info` and `halo_rv_A`. The catalogue contains central candidates, satellite particle candidates, host halo IDs, central/satellite flags, halo particle counts, positions, velocities and host velocities. The expanded satellite reservoir contains 8×10^5 satellite candidates, sufficient to test satellite fractions up to $f_{\text{sat}} \approx 0.15$ at the observed number density.

This catalogue is intentionally not the full AbacusHOD pipeline. It lacks several official ingredients such as the complete AbacusHOD occupation parameterization, DESI cut-sky selection, fibre assignment, and full mock covariance. Its value here is diagnostic: it contains host IDs and velocities in a large public box, allowing us to test how satellite fraction, velocity bias and radial profile interact.

3 METHODS

3.1 Occupation models

We consider two occupation constructions. The first is a SHAM-like reservoir selection: centrals and satellites are independently ranked by a halo proxy and selected to match the target number density and requested satellite fraction. The second is a minimal HOD-like model. Centrals are selected using a noisy mass-rank variable with fixed scatter $\sigma_{\log M} = 0.25$, while satellites are drawn only from hosts whose centrals are occupied and are weighted by host mass with satellite-power index unity. In both cases the total number of galaxies is fixed exactly to the observed number density.

The fitted parameters are the target satellite fraction f_{sat} and the satellite velocity-bias parameter α_s . For a selected satellite, the line-of-sight velocity is

$$v_{\text{sat,los}}^{\text{model}} = v_{\text{host,los}} + \alpha_s (v_{\text{sat,los}} - v_{\text{host,los}}). \quad (2)$$

We then displace galaxies along the z axis to redshift space using the simulation redshift and Planck-like background cosmology (Planck Collaboration et al. 2020).

3.2 Correlation-function measurement

We measure the periodic-box model vectors using `pycorr` with the `Corrfunc` engine (Sinha & Garrison 2020; DESI Collaboration 2024). The model w_p , ξ_0 and ξ_2 are interpolated to the observed bin centres. For rapid grid exploration we use a diagonal heuristic score based on fractional errors and floors:

$$\chi_{\text{heur}}^2 = \sum_i \left[\frac{m_i - d_i}{\sigma_i^{\text{heur}}} \right]^2 + \left[\frac{\ln(\bar{n}_{\text{model}}/\bar{n}_{\text{obs}})}{0.15} \right]^2. \quad (3)$$

Because the number density is matched exactly, the density term is negligible in all successful models. For final observable splits we also compute covariance-weighted scores for w_p , $\xi_0 + \xi_2$, and the full joint vector.

3.3 Satellite radial-profile controls

The first profile control is a direct radial dilution:

$$\mathbf{x}'_{\text{sat}} = \mathbf{x}_{\text{host}} + q (\mathbf{x}_{\text{sat}} - \mathbf{x}_{\text{host}}), \quad (4)$$

with periodic wrapping. This is deliberately phenomenological; it asks how strongly the result changes if satellites are made more extended than the raw reservoir.

The second control is an NFW concentration remapping. We estimate a host radius $R_{200\text{m}}$ from the host particle count and Abacus particle mass, adopt a reference concentration $c_{\text{ref}} = 5$, and compute the normalized radius $y = r/R_{200\text{m}}$. The enclosed NFW fraction is

$$F(y; c) = \frac{\ln(1 + cy) - cy/(1 + cy)}{\ln(1 + c) - c/(1 + c)}. \quad (5)$$

For each satellite we keep its NFW quantile fixed, $F(y; c_{\text{ref}})$, and place it at the corresponding radius in a target profile $c_{\text{sat}} = c_{\text{ref}} c_{\text{ratio}}$. This remapping keeps the satellite direction and velocity residual unchanged while changing the radial concentration in a physically interpretable way.

4 RESULTS

4.1 Baseline HOD and SHAM grids

Table 1 summarizes the main heuristic best fits. The table should be read as a controlled sequence of modelling tests, not as independent final inferences: each row adds one layer of satellite modelling freedom to diagnose the origin of the low- f_{sat} preference.

Figure 1 compares the extended $f_{\text{sat}} - \alpha_s$ grids. The SHAM-like reservoir prefers $f_{\text{sat}} = 0.02$, $\alpha_s = 0.7$, with $\chi_{\text{heur}}^2 \approx 372$. The minimal HOD-like occupation improves the fit and shifts the best point to $f_{\text{sat}} = 0.04$, $\alpha_s = 1.1$, with $\chi_{\text{heur}}^2 = 282.6$. Nevertheless, both models strongly penalize $f_{\text{sat}} \approx 0.10$ – 0.15 , in tension with the order-of-magnitude satellite fractions reported in full DESI One-Percent and BOSS/eBOSS HOD analyses (Yuan et al. 2022, 2024).

The observable split explains the low- f_{sat} preference. The projected clustering $w_p(r_p)$ penalizes high satellite fractions much more strongly than the large-scale multipoles. In the minimal-HOD case, $\xi_0 + \xi_2$ with the full covariance can prefer a somewhat higher satellite fraction, while w_p pulls the joint solution back to low f_{sat} .

4.2 Phenomenological radial dilution

The direct radial-dilution test confirms that the satellite radial profile is a leading systematic. Allowing q in equation (4) improves the minimal-HOD heuristic score from the $q = 1$ baseline $\chi_{\text{heur}}^2 = 282.6$ to $\chi_{\text{heur}}^2 = 222.3$ at $f_{\text{sat}} = 0.06$, $\alpha_s = 1.1$ and $q = 4.0$. Figure 2 shows that increasing q reduces the penalty for higher satellite fractions but does not make $f_{\text{sat}} = 0.10$ – 0.12 competitive.

This behaviour is clearer in Fig. 3. Increasing q systematically reduces the w_p contribution for each satellite fraction. The RSD quadrupole, however, does not respond in the same way. This creates a trade-off: the model can reduce the small-scale projected excess, but it cannot freely move to high f_{sat} without degrading the multipoles.

The covariance-weighted split is more conservative than the heuristic score. For the direct radial-dilution grid, the full joint covariance still selects $f_{\text{sat}} = 0.04$ and $q = 4$, while the heuristic score selects $f_{\text{sat}} = 0.06$. We interpret this not as a contradiction but as evidence that w_p and RSD multipoles are pulling in partially different directions in the simplified model.

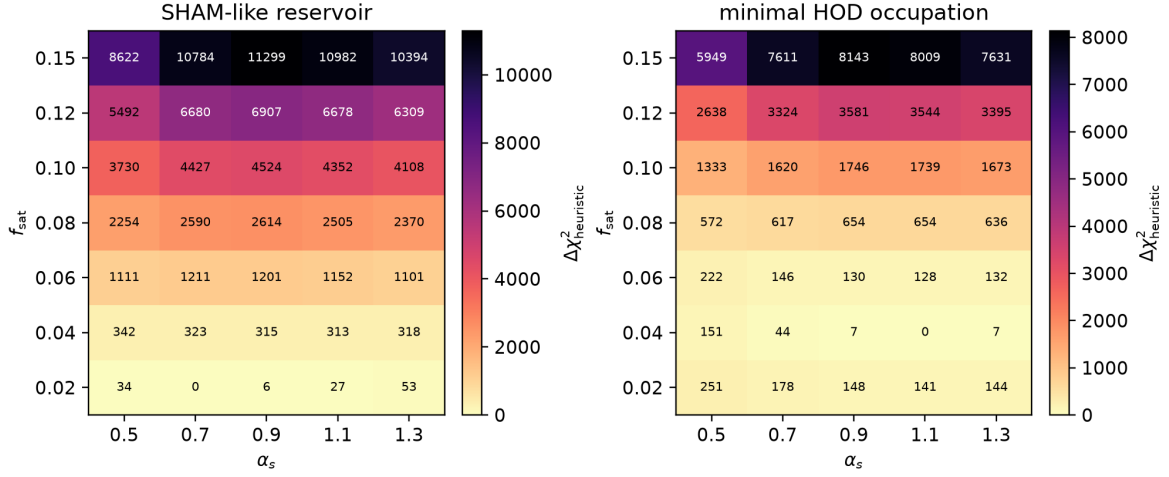


Figure 1. Baseline extended SHAM-like and minimal-HOD grids. Each cell shows $\Delta\chi^2_{\text{heuristic}}$ relative to the best point, minimized over the plotted parameters. The minimal-HOD occupation improves over the SHAM-like reservoir but still favours a low satellite fraction.

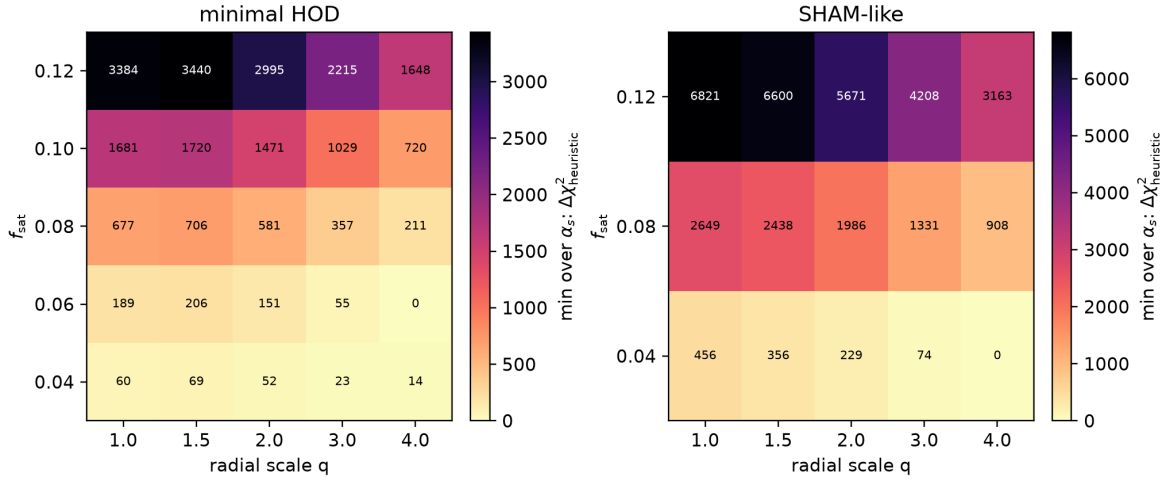


Figure 2. Radial-dilution test. The direct scaling parameter q lowers the small-scale w_p penalty and shifts the heuristic minimal-HOD best fit to $f_{\text{sat}} = 0.06$, but high satellite fractions remain disfavoured.

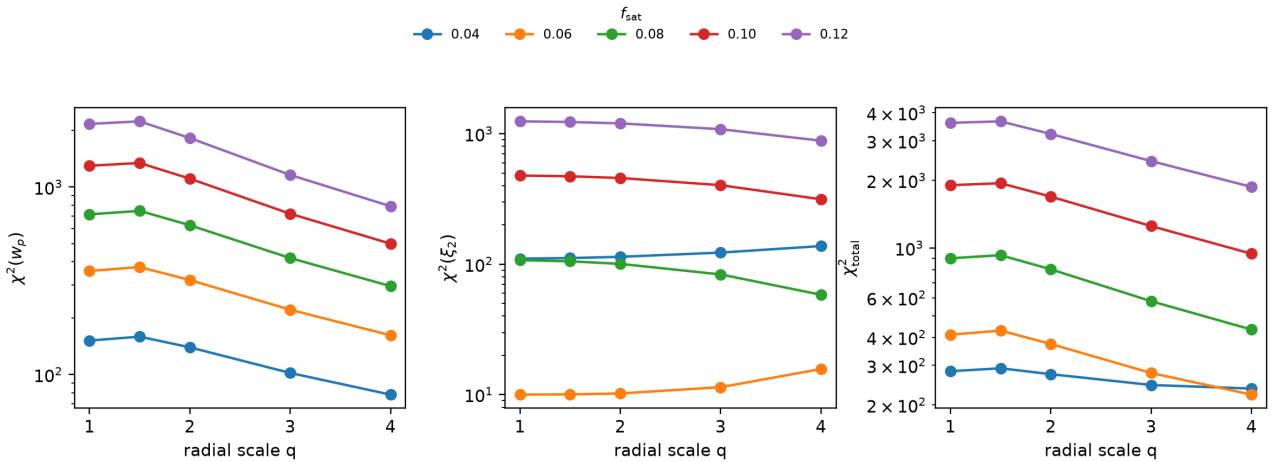


Figure 3. Observable trade-off in the minimal-HOD radial-dilution grid. The main gain from increasing q comes from w_p , while the multipole contribution responds differently. This split is the origin of the joint-fit tension.

Table 1. Summary of the main heuristic best-fit points. The direct radial-dilution parameter q is phenomenological, while the NFW control varies $c_{\text{sat}}/c_{\text{ref}}$ at fixed NFW enclosed-mass quantile.

Model	f_{sat}	α_s	profile parameter	χ^2_{heur}	Interpretation
SHAM-like reservoir	0.02	0.7	none	372	lowest- f_{sat} baseline
minimal HOD	0.04	1.1	none	282.6	occupation layer improves fit
minimal HOD + radial dilution	0.06	1.1	$q = 4.0$	222.3	strong profile sensitivity
minimal HOD + NFW remapping	0.04	1.1	$c_{\text{sat}}/c_{\text{ref}} = 1.3$	274.1	physical concentration control

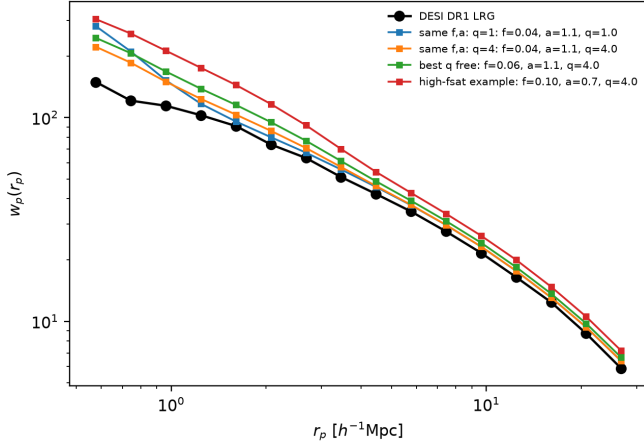


Figure 4. Projected clustering for selected minimal-HOD profile models. The same-parameter $q = 1$ and $q = 4$ curves show directly how radial dilution changes the small-scale w_p mismatch.

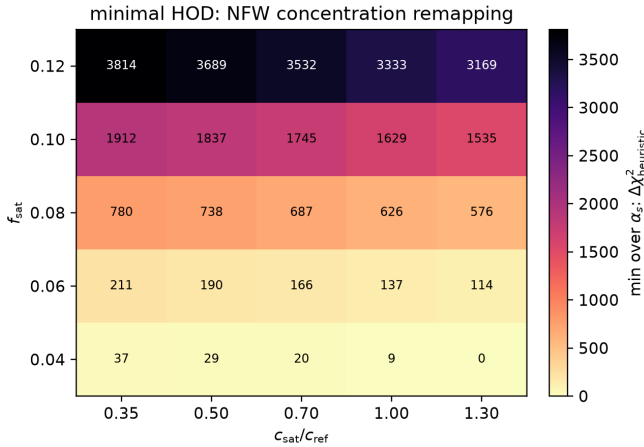


Figure 5. NFW concentration-remapping control. The plotted quantity is $\Delta\chi^2_{\text{heur}}$, minimized over α_s , as a function of target satellite fraction and concentration ratio $c_{\text{sat}}/c_{\text{ref}}$. This physically motivated profile freedom tests whether a standard concentration change can reproduce the large improvement seen in the direct q -dilution experiment.

4.3 NFW concentration-remapping control

The NFW concentration-remapping grid provides a more physical counterpart to the direct q scaling. The best point is $f_{\text{sat}} = 0.04$, $\alpha_s = 1.1$, and $c_{\text{ratio}} = 1.3$ ($c_{\text{sat}} = 6.5$), with $\chi^2_{\text{heur}} = 274.1$. Relative to the $c_{\text{ratio}} = 1$ baseline, $\chi^2_{\text{heur}} = 282.6$, this is only a modest improvement. The remapping changes the satellite radii at fixed NFW quantile rather than applying an arbitrary linear dilation.

The key point is that the NFW remapping does not erase the low- f_{sat} tendency. The best high-satellite rows at $f_{\text{sat}} = 0.10$ and 0.12 have $\chi^2_{\text{heur}} = 1809$ and 3443.4 , respectively, still far above the low- f_{sat} solutions. Thus, within this constrained NFW-family control, varying concentration alone is not sufficient to recover the $f_{\text{sat}} \approx 0.10$ – 0.15 region while also matching $\xi_0 + \xi_2$.

5 DISCUSSION

The strongest conclusion is not that DESI LRG clustering generically requires a very low satellite fraction. Rather, in this public full-box compact-mock setup, the inferred f_{sat} is highly sensitive to the satellite radial profile used by the simplified occupation model. A model with too much one-halo projected power is driven toward low f_{sat} by $w_p(r_p)$, even if the redshift-space multipoles could accommodate a larger satellite contribution.

The main empirical signature is therefore not a uniform preference across the whole clustering vector. The low- f_{sat} preference is driven by w_p , not by the full clustering vector uniformly. In the split fits, small-scale projected clustering carries the strongest penalty against high satellite fractions, while $\xi_0 + \xi_2$ can move in a different direction once velocity bias and profile changes are allowed. This observable dependence is the reason we interpret the result as a profile-systematics diagnostic rather than a direct physical measurement of the DESI LRG satellite fraction.

This conclusion also explains why the result should not be read as a contradiction of full AbacusHOD analyses. DESI One-Percent AbacusHOD fits find $f_{\text{sat}} \approx 0.10$ – 0.14 , consistent with broader BOSS/eBOSS LRG HOD expectations (Yuan et al. 2022, 2024; Zhai et al. 2017). Those analyses marginalize over richer occupations, assembly-bias or velocity-bias freedom, cleaned haloes, DESI-like selection and more realistic covariance treatment. Our lower values are diagnostic preferences from a compact public-box reservoir. The discrepancy is best read as evidence that an over-concentrated or otherwise mismatched satellite reservoir can force simplified fits toward artificially low f_{sat} .

The practical implication is that resource-limited mock analyses should split w_p from $\xi_0 + \xi_2$ and introduce at least one satellite-profile control before interpreting f_{sat} astrophysically. The remaining limitations are important: the covariance is a 96-region jackknife, the box is periodic, fibre assignment and imaging systematics are absent, and the minimal-HOD layer is deliberately simpler than AbacusHOD. These choices are appropriate for a diagnostic Letter but not for a definitive DESI HOD inference.

6 CONCLUSIONS

We have used a public Abacus full-box compact catalogue to study the sensitivity of DESI LRG HOD inference to satellite fraction, velocity bias and satellite radial profile. This is not a replacement for

full DESI AbacusHOD inference, but a diagnostic study of satellite-profile systematics. The main results are:

- (i) SHAM-like and minimal-HOD samples both prefer low f_{sat} in the baseline $w_p + \xi_0 + \xi_2$ fit, with the minimal-HOD grid favouring $f_{\text{sat}} = 0.04$.
- (ii) The low- f_{sat} preference is driven by small-scale w_p , not by the full clustering vector uniformly.
- (iii) Direct radial dilution improves the heuristic minimal-HOD fit from $\chi^2 = 282.6$ to $\chi^2 = 222.3$ and shifts the best point to $f_{\text{sat}} = 0.06$.
- (iv) NFW concentration remapping gives only a modest improvement, so ordinary concentration changes alone do not make $f_{\text{sat}} = 0.10\text{--}0.15$ competitive.

The next valuable step is a full joint inference of f_{sat} , α_s and satellite profile using an official AbacusHOD or equivalent model with DESI-like selection, fibre assignment and mock covariance.

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DATA AVAILABILITY

The analysis scripts, lightweight summary tables, figure-generation products and manuscript source are available at <https://github.com/billkang-x/desi-lrg-satellite-profile-systematics>. Large external catalogues and regenerated binary products are not included in the repository. The AbacusSummit input products are publicly available from the AbacusSummit data release, and the DESI clustering measurements used here are public DESI DR1 LRG measurements or derived products from the same data vector.

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